

OOPs Concepts in JavaScript - Complete Notes

OOPs in JavaScript has four main principles: Inheritance, Polymorphism, Encapsulation, and Abstraction.

1. INHERITANCE

Definition: Allows one class to acquire properties and methods of another class.

Achieved using 'extends' and 'super()'.

2. POLYMORPHISM

Definition: Same method behaves differently based on the object. Achieved by method overriding.

3. ENCAPSULATION

Definition: Binding data and methods into one unit and hiding data. Achieved using private fields (#) and getters/setters.

4. ABSTRACTION

Definition: Hiding complex implementation and showing only necessary info.

Summary Table:

Inheritance - Reusability (extends, super())

Polymorphism - Flexibility (Method Overriding)

Encapsulation - Data Security (Private fields, Getters/Setters)

Abstraction - Simplicity (Private methods, Interfaces)